

Toward a theory of multi-objective learning

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Geelon So | EnCORE Collaboration Workshop





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Motivating example: self-driving car

Goal: train a model for a self-driving car.



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The learning problem

Directly optimizing R is not possible since we only have sample access to it.

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What we might care about:

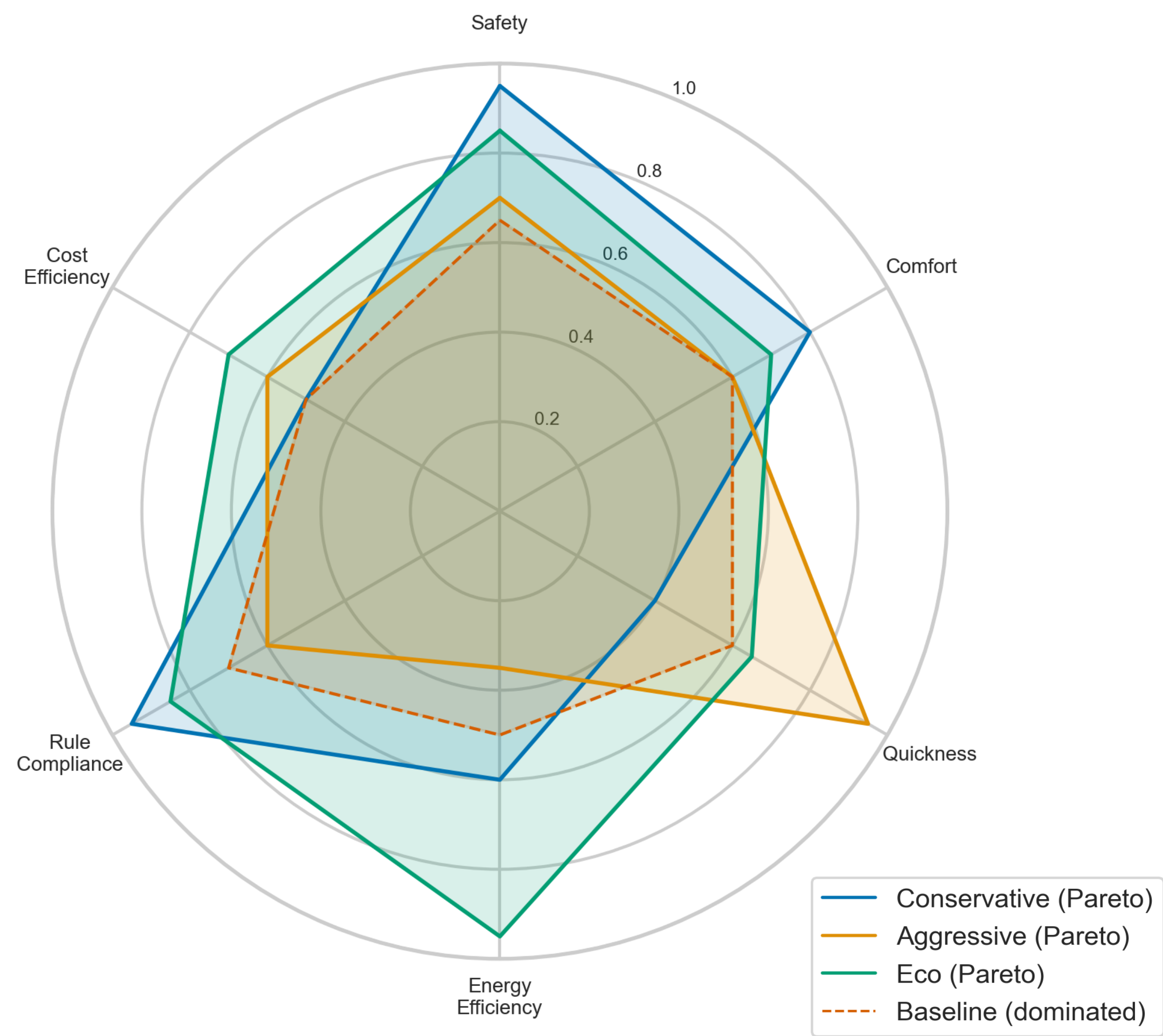
- Safety
- Efficiency
- Comfort
- [many more]

Multi-objective risk minimization

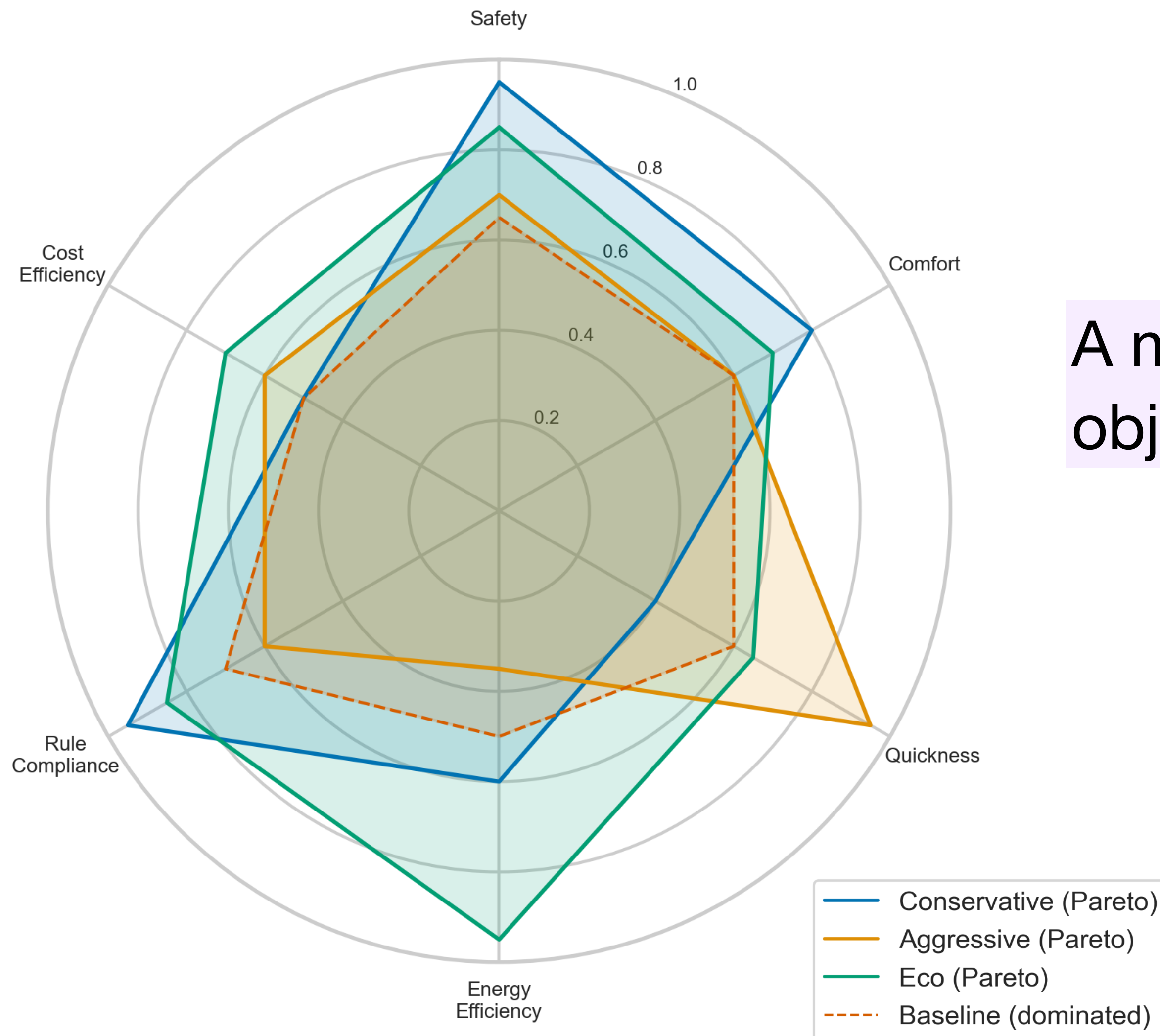
$$\min_{f \in \mathcal{F}} \underbrace{\mathbf{R}(f) \equiv (R_1(f), \dots, R_K(f))}$$

Now, we care about many types of risks $\mathbf{R}(f)$.

Solution concept: Pareto optimality

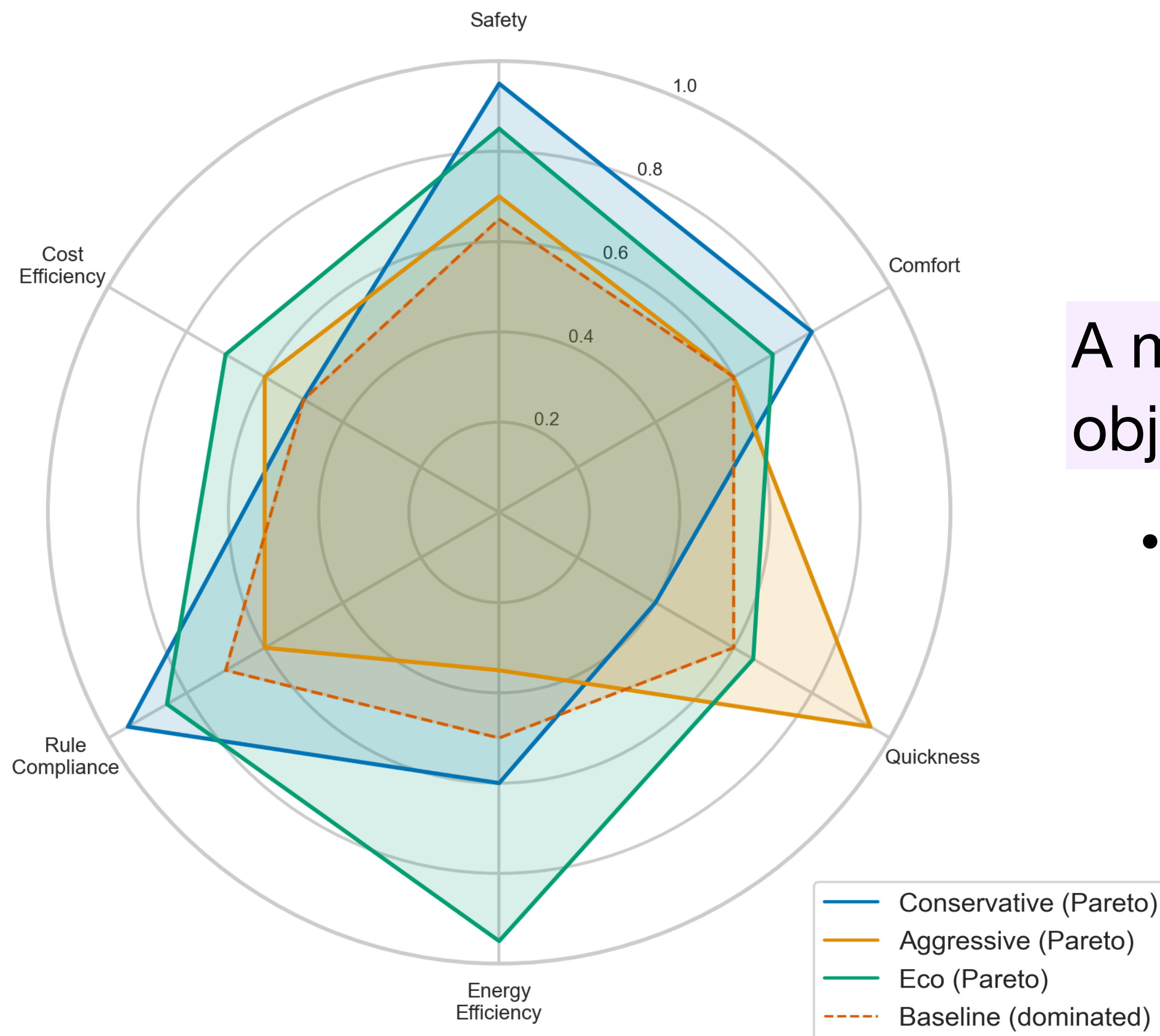


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- The specific type of *trade-off* is not usually known beforehand.

Question: How much data is needed to learn all Pareto optimal models?

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- What if we already something about how to solve the individual tasks?

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- $f_k^\star$ is the Bayes-optimal model minimizing R_k

A negative result

Theorem. Let $\mathcal{F}$ be a model class. To ε -learn all Pareto optimal models, we need:

$$\tilde{\Theta} \left(\frac{\text{VC}(\mathcal{F}) \cdot K}{\varepsilon^2} \right) \text{ samples,}$$

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Problem: loss functions such as the zero-one loss can be “uninformative”.

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Importantly, the label sample complexity does not the complexity of the joint class $\mathcal{F}$ in which the good trade-offs are possible.

Takeaways

- Multi-objective learning (MOL) problems are ubiquitous in practice.
- Learning good trade-offs can be much harder than solving the individual tasks.
- Structure in loss/feedback important for efficient multi-objective generalization.

Thanks!

**On the sample complexity of semi-supervised
multi-objective learning**

<https://arxiv.org/abs/2508.17152>